

## Unit 6.2 - Vectors, Section 12.2 p798

Note Title

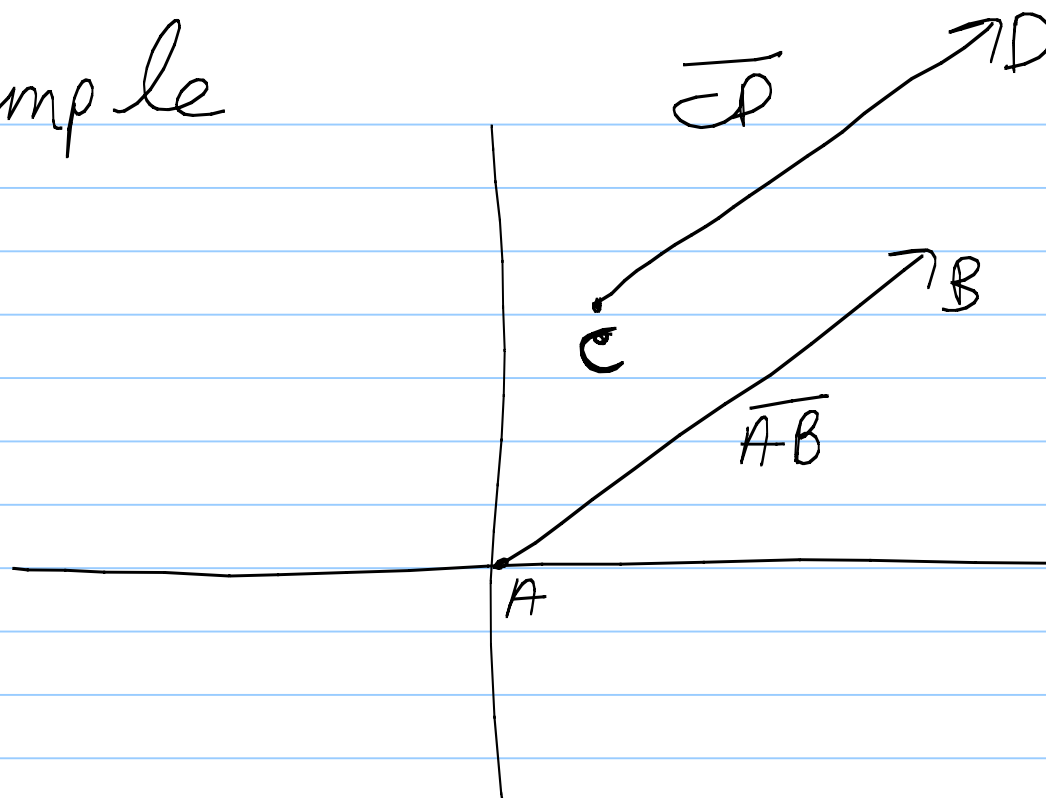
2012/05/20

Definition: A vector is a quantity that has magnitude and direction.

Remark: A vector does not have a fixed position.

Notation: The vector from point A to B is written as  $\vec{AB}$ .

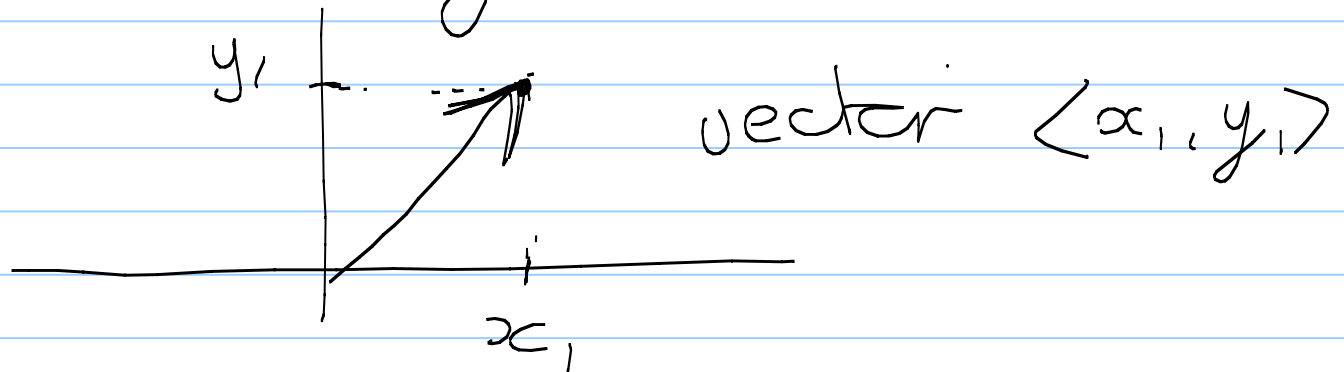
Example



## Definition

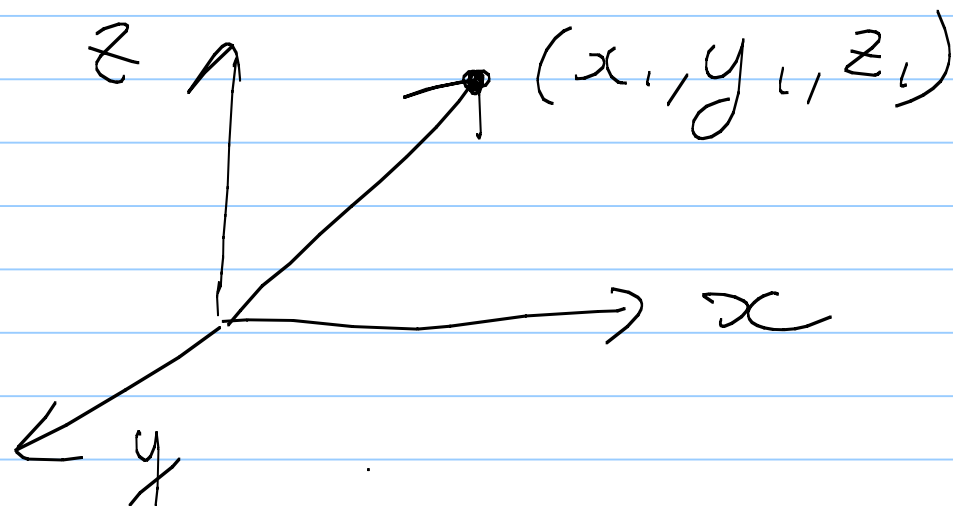
In  $\mathbb{R}^2$ :

A vector from the origin  $(0,0)$  to the point  $(x_1, y_1)$  is written  $\langle x_1, y_1 \rangle$



In  $\mathbb{R}^3$ :

A vector from the origin  
 $(0, 0, 0)$  to the point  
 $(x_1, y_1, z_1)$  is written as  
 $\langle x_1, y_1, z_1 \rangle$



Example

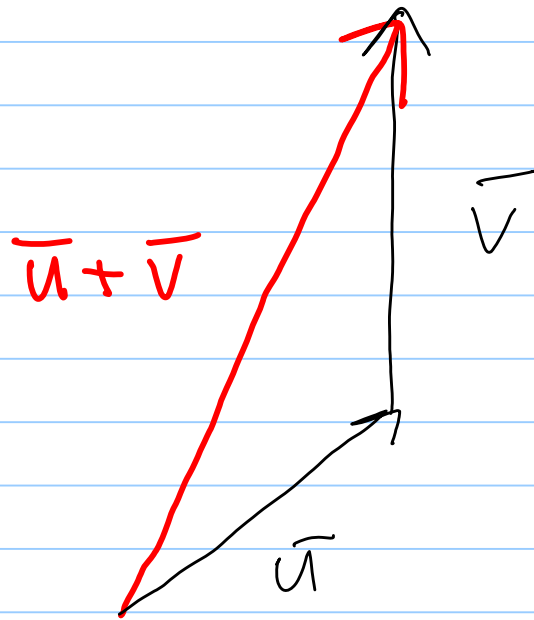
Sketch

the vectors  $\langle -1, 2 \rangle$  and

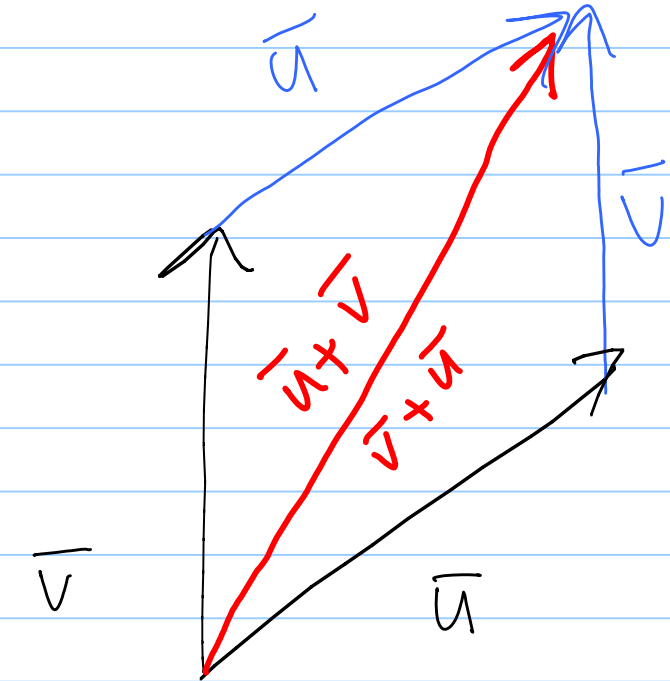
$\langle 1, 2, 3 \rangle$

## Def of vector addition

If  $\vec{u}$  and  $\vec{v}$  are vectors positioned so the initial point of  $\vec{v}$  is at the terminal point of  $\vec{u}$ , then the sum  $\vec{u} + \vec{v}$  is the vector from the initial point of  $\vec{u}$  to the terminal point of  $\vec{v}$ .



Triangle  
Law



Parallelogram  
Law

## Def of scalar multiplication

If  $c$  is a scalar and  $\vec{v}$  is a vector, then the scalar multiple  $c\vec{v}$  is the vector whose length is  $|c|$  times the length of  $\vec{v}$ .

- If  $c > 0$  then  $c\vec{v}$  has same direction
- If  $c < 0$  then  $c\vec{v}$  is in the

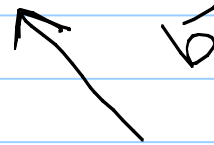
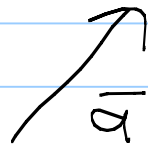
opposite direction

• If  $c=0$ , Then  $c\vec{v}=\vec{0}$   
where  $\vec{0}$  is a zero vector.

NB

Note: Two non zero vectors  
are parallel if they are  
scalar multiples of one  
another

Example Vector  $\vec{a}$  and  $\vec{b}$   
are given in the sketch



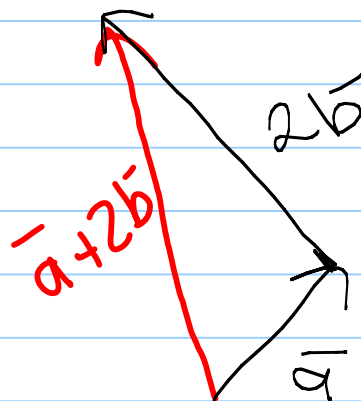
Sketch

①  $\vec{a} + 2\vec{b}$

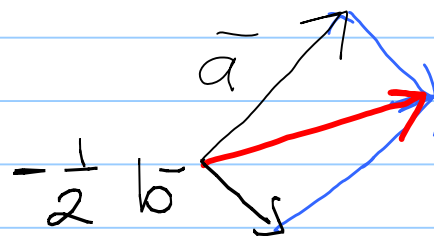
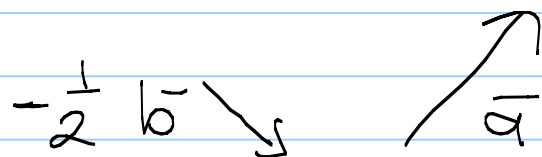
②  $\vec{a} - \frac{1}{2}\vec{b}$

Sol:

①



②



## Definition

If  $A(x_1, y_1)$  and  $B(x_2, y_2)$  are points in  $\mathbb{R}^2$  then

$$\overline{AB} = \langle x_2 - x_1, y_2 - y_1 \rangle$$

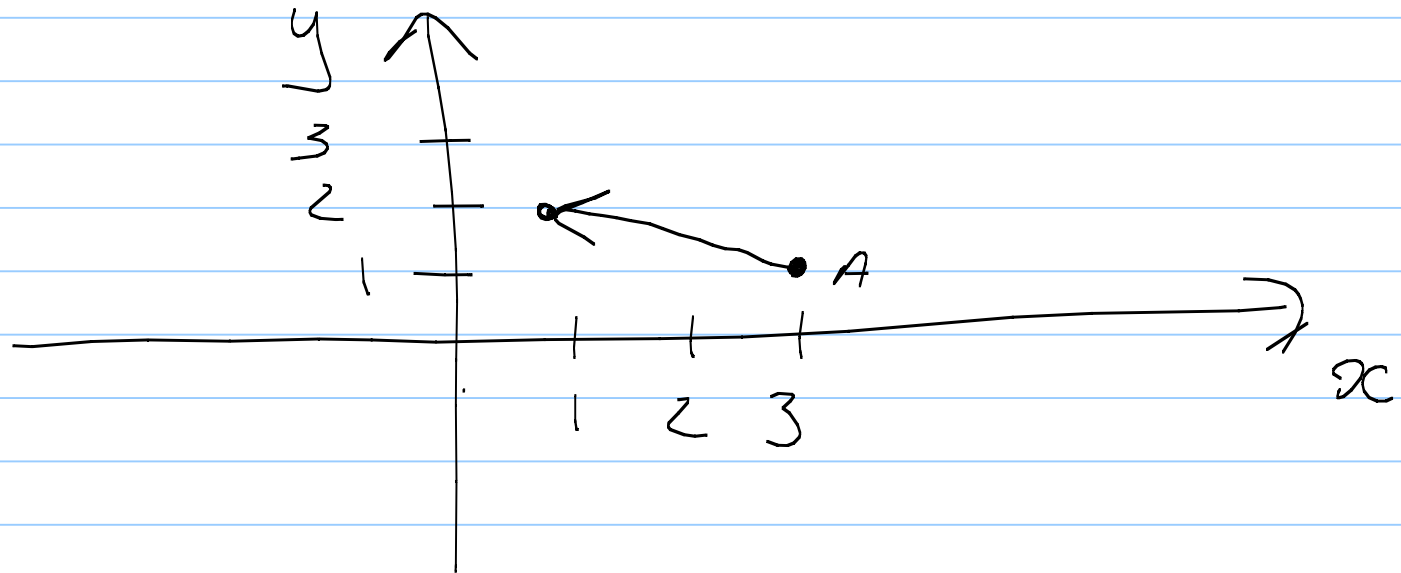
If  $A(x_1, y_1, z_1)$  and  $B(x_2, y_2, z_2)$  are points in  $\mathbb{R}^3$  then

$$\overline{AB} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Example  $A(3,1)$  and  $B(1,2)$

Find  $\overrightarrow{AB}$  and Sketch in  $\mathbb{R}^2$ .

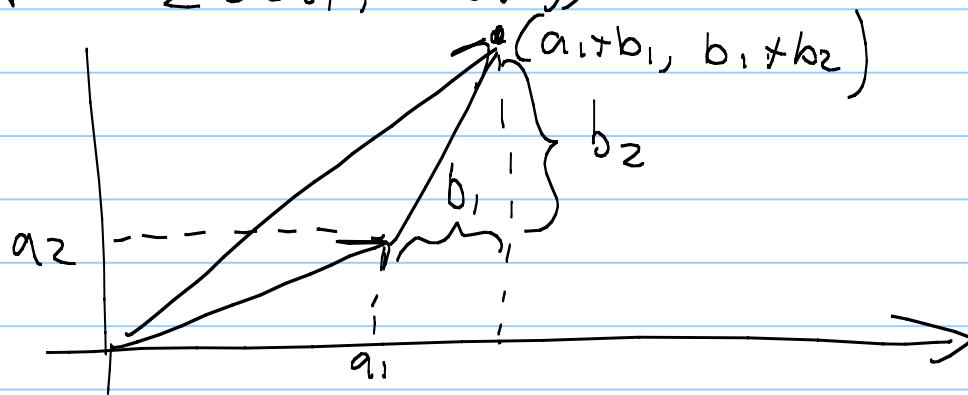
$$\overrightarrow{AB} = \langle 1-3, 2-1 \rangle = \langle -2, 1 \rangle$$



Definition: of Vector addition, subtraction  
and scalar multiplication

If  $\vec{a} = \langle a_1, a_2 \rangle$  and  $\vec{b} = \langle b_1, b_2 \rangle$

- $\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2 \rangle$
- $\vec{a} - \vec{b} = \langle a_1 - b_1, a_2 - b_2 \rangle$
- $c\vec{a} = \langle ca_1, ca_2 \rangle$



In  $\mathbb{R}^3$ :

If  $\vec{a} = \langle a_1, a_2, a_3 \rangle$  and  $\vec{b} = \langle b_1, b_2, b_3 \rangle$

Then

- $\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$

- $\vec{a} - \vec{b} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$

- $c\vec{a} = \langle ca_1, ca_2, ca_3 \rangle$

Example Given  $A(2, -1, 0)$ ,  $B(-2, 4, 1)$   
and  $C(1, 2, -2)$

Find  $2\overrightarrow{AB} - 3\overrightarrow{CA}$

$$\begin{aligned}\overrightarrow{AB} &= \langle -2 - 2, 4 - (-1), 1 - 0 \rangle \\ &= \langle -4, 5, 1 \rangle\end{aligned}$$

$$\begin{aligned}\overrightarrow{CA} &= \langle 2 - 1, -1 - 2, 0 - (-2) \rangle \\ &= \langle 1, -3, 2 \rangle\end{aligned}$$

$$2\overline{AB} - 3\overline{CA}$$

$$= 2\langle -4, 5, 1 \rangle - 3\langle 1, -3, 2 \rangle$$

$$= \langle -8, 10, 2 \rangle - \langle 3, -9, 6 \rangle$$

$$= \langle -11, 19, -4 \rangle$$

# Properties of vector addition, subtraction and scalar multiplication

$$1. \quad \vec{a} + \vec{b} = \vec{b} + \vec{a}$$

$$2. \quad \vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$$

$$3. \quad \vec{a} + \vec{0} = \vec{a}$$

$$4. \quad \vec{a} + (-\vec{a}) = \vec{0}$$

$$5. \quad c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$$

$$6. (c + d) \bar{a} = c\bar{a} + d\bar{a}$$

$$7. (cd) \bar{a} = c(d\bar{a})$$

$$8. 1\bar{a} = \bar{a}$$

## Definition

Vectors  $\vec{a}$  and  $\vec{b}$  are parallel if there exists a  $c \in \mathbb{R}$  such that

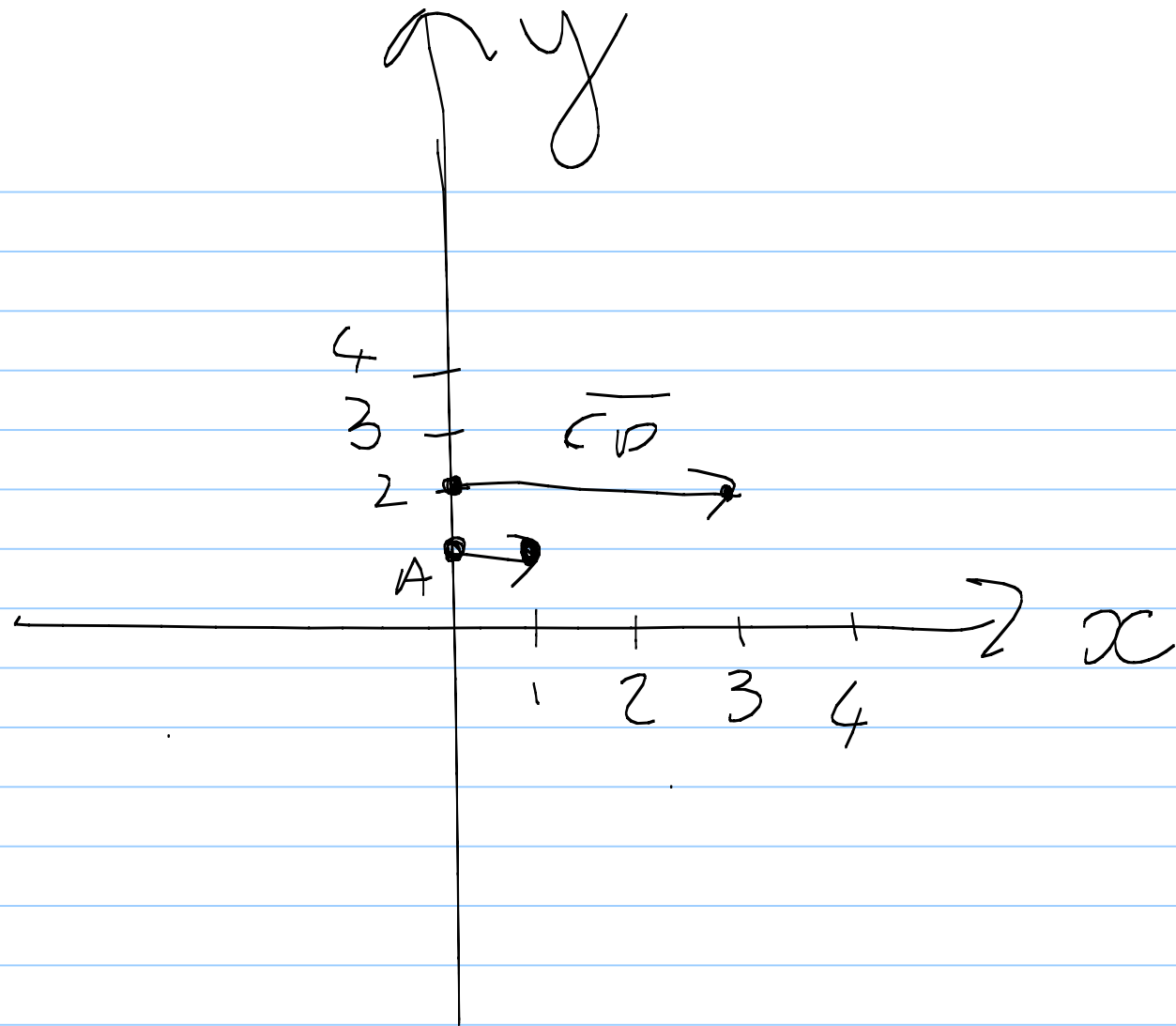
$$\vec{a} = c\vec{b}$$

Example. If  $A(0,1)$ ,  $B(1,1)$ ,  $C(0,2)$  and  $D(3,2)$  are points, use a sketch and the definition to show that  $\overline{AB}$  and  $\overline{CD}$  are parallel.

Sol:

$$\overline{AB} = \langle 1, 0 \rangle$$

$$\overline{CD} = \langle 3, 0 \rangle = 3 \langle 1, 0 \rangle$$



## Definition

The length / norm of a vector  $\vec{a}$  is defined as

$$\begin{aligned} |\vec{a}| &= |\langle a_1, a_2 \rangle| \\ &= \sqrt{a_1^2 + a_2^2} \end{aligned}$$

Definition: A unit vector is a vector of length one.

Example:  $\vec{a} = \langle 1, 1, 1 \rangle$  is not a unit vector in  $\mathbb{R}^3$ .

Sol:  $|\vec{a}| = |\langle 1, 1, 1 \rangle|$

$$= \sqrt{1^2 + 1^2 + 1^2}$$
$$= \sqrt{3}$$

Example:  $\bar{a} = \langle \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \rangle$  is a unit vector in  $\mathbb{R}^2$ .

Sol:  $|\bar{a}|$

$$= \sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2}$$

$$= \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1}$$

$$= 1 \quad \square$$

Remark: If  $\vec{a} \neq \vec{0}$  then  
 $\frac{1}{|\vec{a}|} \vec{a}$  is a unit vector in the  
same direction as  $\vec{a}$ .

Example: Find a vector of length  
2 in the opposite direction as  
 $\vec{a} = \langle -3, 4 \rangle$ .

Solution

$$|\bar{a}| = \sqrt{(-3)^2 + (4)^2} = \sqrt{25}$$

The vector is  $-\frac{2}{|\bar{a}|} \bar{a}$

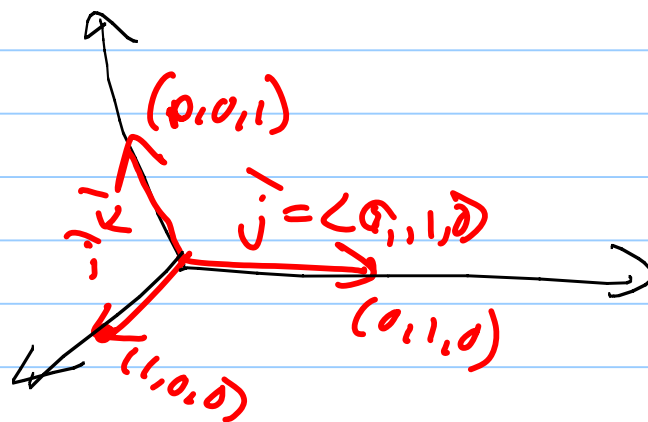
$$= \frac{-2}{\sqrt{25}} \cdot (-3, 4)$$

Definition

The standard basis vectors of

•  $\mathbb{R}^2$  is  $\vec{i} = \langle 1, 0 \rangle$  and  $\vec{j} = \langle 0, 1 \rangle$

•  $\mathbb{R}^3$  is  $\vec{i} = \langle 1, 0, 0 \rangle$ ,  $\vec{j} = \langle 0, 1, 0 \rangle$  and  
 $\vec{k} = \langle 0, 0, 1 \rangle$



Remark: You can write any vector in  $\mathbb{R}^2$  or  $\mathbb{R}^3$  in terms of the standard basis vectors.

Example

$$\langle a, b, c \rangle = \langle a, 0, 0 \rangle + \langle 0, b, 0 \rangle + \langle 0, 0, c \rangle$$

$$= a \langle 1, 0, 0 \rangle + b \langle 0, 1, 0 \rangle + c \langle 0, 0, 1 \rangle$$

$$= a \vec{i} + b \vec{j} + c \vec{k}.$$

Example  $\vec{a} = \langle 1, 2, -5 \rangle$  ,  $\vec{b} = \langle 0, 1, 2 \rangle$

Find  $\vec{a} - \vec{b}$  write your answer in terms of  $\vec{i}$ ,  $\vec{j}$ ,  $\vec{k}$

Sol:

$$\vec{a} - \vec{b}$$

$$= \langle 1, 2, -5 \rangle - \langle 0, 1, 2 \rangle$$

$$= \langle 1, 1, -7 \rangle$$

$$= \vec{i} + \vec{j} - 7\vec{k}$$